SOFTWARE DEVELOPMENT • RefView Window/Level Display Transform

Exercise 3 — Stop Asking AI to Do Everything at Once

FACILITATOR EDITION — DO NOT DISTRIBUTE TO PARTICIPANTS

Participant task: Before changing the implementation, use AI to enumerate assumptions, ambiguities, boundary conditions, invalid-input behavior, numerical decisions, and discriminating tests. Ask: "What information would you want clarified before finalizing this component?"

LOCKED ACCEPTANCE CRITERIA — unchanged across Exercises 1–5

- Correctly maps values below, inside, and above the window.
- Always returns an integer from 0–255.
- Produces deterministic results.
- Defines predictable invalid-window behavior.
- Includes normal, boundary, and invalid-input tests.

FACILITATOR RULE — release answers only when the team identifies the corresponding question

Do not hand out this table. Reward discovery of missing information, not guessing the hidden answer.

Question the team discovers	Release this clarification
How are .5 values rounded?	Round half away from zero.
What numeric types may inputs use?	Pixel value, center, and width may be integer or floating point.
What about NaN or infinity?	Invalid input; return an explicit error.
Is width exactly 1 valid?	No. Width must be greater than 1.
Can output exceed 0–255 internally?	Clamp after calculation and rounding.
Is display inversion required?	No. Low values are black; high values are white.

WHAT TO LISTEN FOR / INTENDED INSIGHT

- Strong teams ask targeted questions before coding rather than asking AI for another complete solution.
- Listen for separation of interface contract, numeric policy, error policy, and test design.
- The improvement comes from iterative information discovery—not from a longer one-shot prompt.

Exercise 3 — Stop Asking AI to Do Everything at Once

FACILITATOR EDITION — DO NOT DISTRIBUTE TO PARTICIPANTS

Participant task: Decompose the science: measurands, control, held-constant variables, independence, replication, uncertainty, false-success mechanisms, and falsifying evidence. Ask AI what information is still missing.

LOCKED ACCEPTANCE CRITERIA — unchanged across Exercises 1–5

- Quantitatively evaluates noise, spatial resolution, and CT-number bias.
- Includes an appropriate matched control.
- Addresses repeatability and measurement uncertainty.
- Identifies important confounders.
- Defines objective support/fail criteria: noise ≥30% reduction; resolution degradation ≤5%; absolute introduced bias ≤3 HU.

FACILITATOR RULE — release answers only when the corresponding issue is raised

The exercise tests whether teams discover missing information themselves.

Question the team discovers	Release this clarification
Are processed/control images separate scans?	No. They are paired from the same acquisition.
Can multiple slices be measured?	Yes.
Can multiple ROIs be used?	Yes, but correlated measurements are not independent samples.
May ROI location change?	No. Use matched locations.
Required resolution metric?	MTF50.
How is CT-number bias calculated?	Processed minus corresponding unprocessed; nominal values are an additional reference.
Must uncertainty be quantified?	Yes: confidence intervals or another justified representation.

WHAT TO LISTEN FOR / INTENDED INSIGHT

- Strong plans distinguish observations from independent experimental units.
- Listen for pairing, matched ROIs, replication, uncertainty, and confounder control.
- Decomposition should expose what the team does not yet know—not just reorganize the answer.

PRODUCT DEVELOPMENT • RefView Auto-Window Capability

Exercise 3 — Stop Asking AI to Do Everything at Once

FACILITATOR EDITION — DO NOT DISTRIBUTE TO PARTICIPANTS

Participant task: Decompose intended behavior, I/O, performance, determinism, fallback, override, abnormal inputs, technical risk, and verification evidence. Ask AI what is missing before implementation and objective verification.

LOCKED ACCEPTANCE CRITERIA — unchanged across Exercises 1–5

- Objectively testable functional requirements.
- Measurable performance requirements.
- Defined abnormal/unsupported-input behavior.
- Significant technical risks and mitigations.
- Verification evidence and objective pass/fail criteria for every requirement, including normal and edge cases.

FACILITATOR RULE — release answers only when asked through a discovered question

Do not provide the table as a handout.

Question the team discovers	Release this clarification
What is the fallback?	Center=40, Width=400.
Too many invalid pixels?	If fewer than 95% are valid numeric values, use fallback.
Calculated width <2?	Use fallback.
Rerun after manual adjustment?	No. Run only when the image initially opens.
May it modify pixel data?	No. Display parameters only.
Must identical input match?	Yes, deterministic.
What hardware defines 200 ms?	The fictional RefView reference workstation.
Is “looks good” acceptable?	No. Use expected outputs or tolerances.

WHAT TO LISTEN FOR / INTENDED INSIGHT

- Strong teams convert broad capability language into an explicit contract and evidence plan.
- Listen for fallback thresholds, run timing, determinism, non-modification of pixels, and objective verification.
- Missing-information discovery should precede document polishing.